Run One sjSDM Predictor Structure Across Selected Folds

Description

Applies one controlled predictor structure to deterministic selected-candidate folds and returns labelled predictions and diagnostics.

Usage

```
run_sjsdm_predictor_structure_folds(
  predictor_structure = NULL,
  data_assignments = NULL,
  data_selected_candidate = NULL,
  data_community_matrix = NULL,
  data_abiotic_wide = NULL,
  data_coords_projected = NULL,
  data_sample_ids = NULL,
  config_model_fitting = NULL,
  config_data_processing = NULL,
  model_formula = NULL,
  device = "gpu",
  seed = 900723L,
  prepare_fold_function = prepare_sjsdm_cross_validation_fold,
  fit_candidate_function = fit_sjsdm_cross_validation_candidate,
  runner_function = run_sjsdm_selected_candidate_folds,
  predict_function = predict_sjsdm_probability_matrix
)
```

Arguments

`predictor_structure`	Structure passed to [build_sjsdm_predictor_comparison_structure()].

`data_assignments`, `data_selected_candidate`, `data_community_matrix`	Cross-validation assignments, selected regularization, and response matrix.
`data_abiotic_wide`, `data_coords_projected`, `data_sample_ids`	Abiotic predictors, projected coordinates, and aligned sample metadata.
`config_model_fitting`, `config_data_processing`	Model-fitting and data-processing configurations used by fold preparation.
`model_formula`	Full abiotic formula used when abiotic predictors are enabled.
`device`	Fitting device passed to [fit_sjsdm_regularization_candidate()].
`seed`	Base selected-fold seed passed to [run_sjsdm_selected_candidate_folds()].
`prepare_fold_function`, `fit_candidate_function`, `runner_function`	Injectable functions for fold preparation, candidate fitting, and selected fold orchestration.
`predict_function`	Injectable held-out prediction function.

Details

Regularization and assignments are fixed by the supplied reference inputs. The helper changes predictor inclusion only. Intercept-only and spatial-only structures use an environmental intercept because sjSDM requires an environmental component. The no-associations structure retains abiotic and spatial predictors while restricting the biotic covariance to its diagonal.

Value

Named list containing labelled 'data_predictions' and 'data_diagnostics'.

Examples

```
## Not run:

run_sjsdm_predictor_structure_folds(

  predictor_structure = "abiotic_only",

  data_assignments = data_assignments,

  data_selected_candidate = data_selected_candidate,

  data_community_matrix = data_community_matrix,
```

```
  data_abiotic_wide = data_abiotic_wide,
  data_coords_projected = data_coords_projected,
  data_sample_ids = data_sample_ids,
  config_model_fitting = config_model_fitting,
  config_data_processing = config_data_processing,
  model_formula = model_formula
)

## End(Not run)
```